

NPRE 332 (borrowed from ME 231)

Fall 2003

Phase Diagrams & Casting Questions

Analysis of Results and Points for Discussion

1. Plot *two* of the cooling curves (temperature vs. time) measured during lab (one for a pure metal and one for any alloy that is not the eutectic). Be sure to report the composition of each sample. Briefly explain how the values for the liquidus and solidus temperatures are found from the cooling curves, and indicate those points on the cooling curves. Explain why the shape of the alloy cooling curve is different for the pure metal.
2. Four microstructures (0 wt% Bi, 30 wt% Bi, 50 wt.% Bi, and 55 wt.% Bi) have been posted on the web site. Plot the four Bi-Sn cooling curves that correspond the four microstructures. Explain the formation of these four microstructures using the cooling curves. Be sure to report the composition of each sample and microstructure.
3. Collect all of the experimental data of liquidus, solidus, and solvus (where appropriate) temperatures for the Sn-Bi system as a function of composition. There are 16 compositions available. Report these data in a table. How do the measured melting temperatures of Bi and Sn compare with published values?
4. Using all the cooling curve data, compare your diagram with the published phase diagram for the Sn-Bi system by plotting experimental data directly on top of published phase diagram. Label each region. List at least 3 sources of variability in **your** measurements and discuss their effects (over/under predicted temperatures). How accurately does your experimental data allow you to locate the eutectic temperature and composition?
5. You were given three pages showing macrographs of several aluminum cast specimens. The actual specimens were examined in laboratory. Fill in the pertinent information (composition, mold type, and superheat) on these pages. Include the pages as an appendix to your lab report (with the blanks filled in).
6. How does addition of Cu and Ti affect the grain structure of the cast Al samples? Which is more effective? What is the effect of heating the mold? How about shaking the mold?
7. What factors make the columnar zone larger relative to the equiaxed zone? What can be done to affect the size of the chill zone adjacent to the mold? Which of the cast specimens had the most uniform macrostructure? Why? What are the advantages/disadvantages of this?